



URBAN HIGHWAYS ARE BARRIERS TO SOCIAL TIES

HIGHWAYS ARE BARRIERS

Study examines:

- Social ties across 50 U.S. cities
- Highways as structural dividers

Measuring it with:

- Twitter mutual follows as proxy for social ties



How do highways reduce social connectivity?

KEY FINDINGS

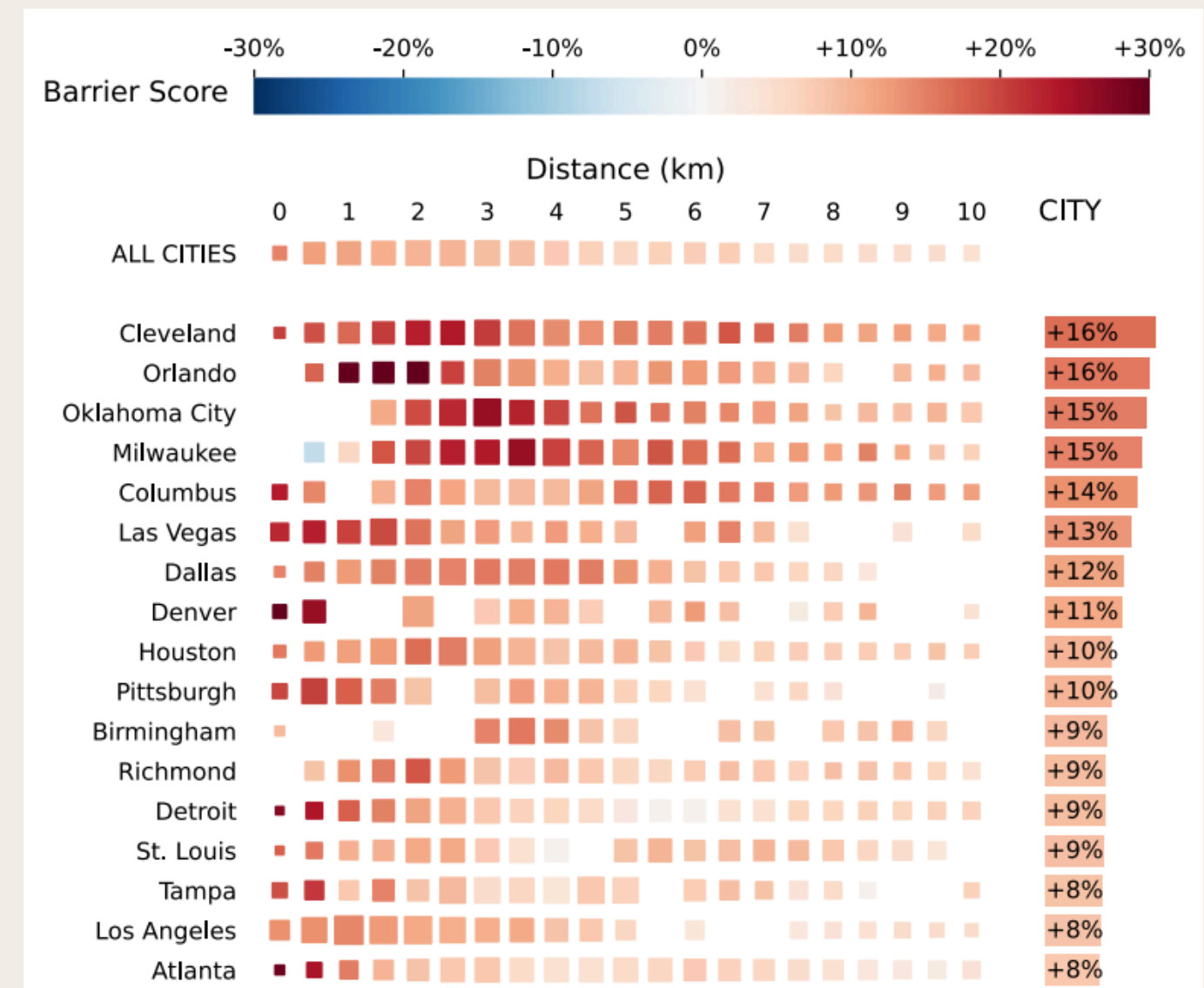
1

HIGHWAYS REDUCE SOCIAL CONNECTIVITY ACROSS ALL 50 MAJOR US CITIES

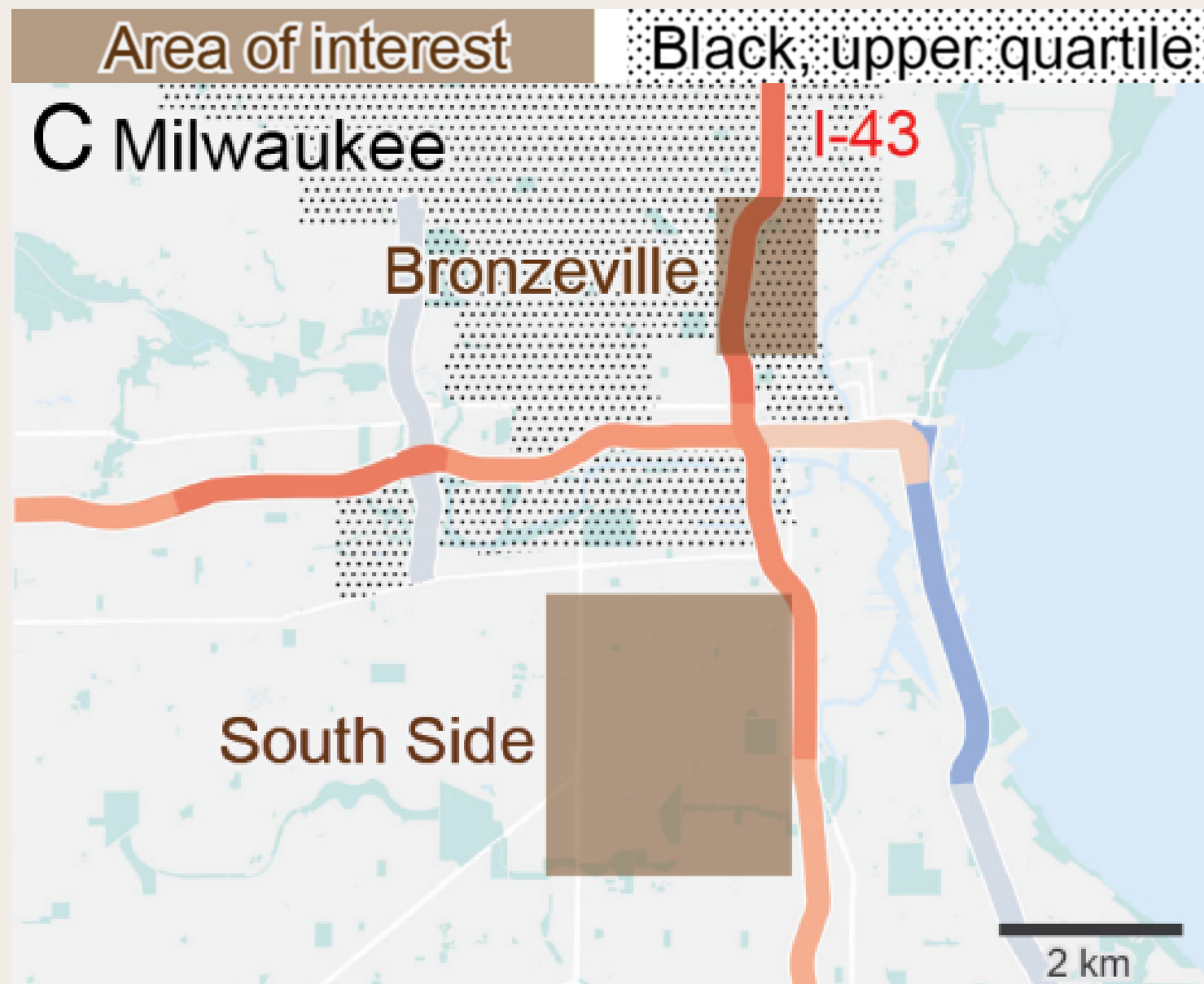
Positive **Barrier Scores** in all 50 cities

Up to 16% fewer ties across highways

Strongest effect at 1-5 km distances



KEY FINDINGS



2

HIGHWAYS USED FOR RACIAL SEGREGATION STILL PRODUCE THE STRONGEST BARRIER EFFECTS

Segregation-era highways show **highest scores**

Barriers align with Black-majority neighborhoods

BARRIER SCORE

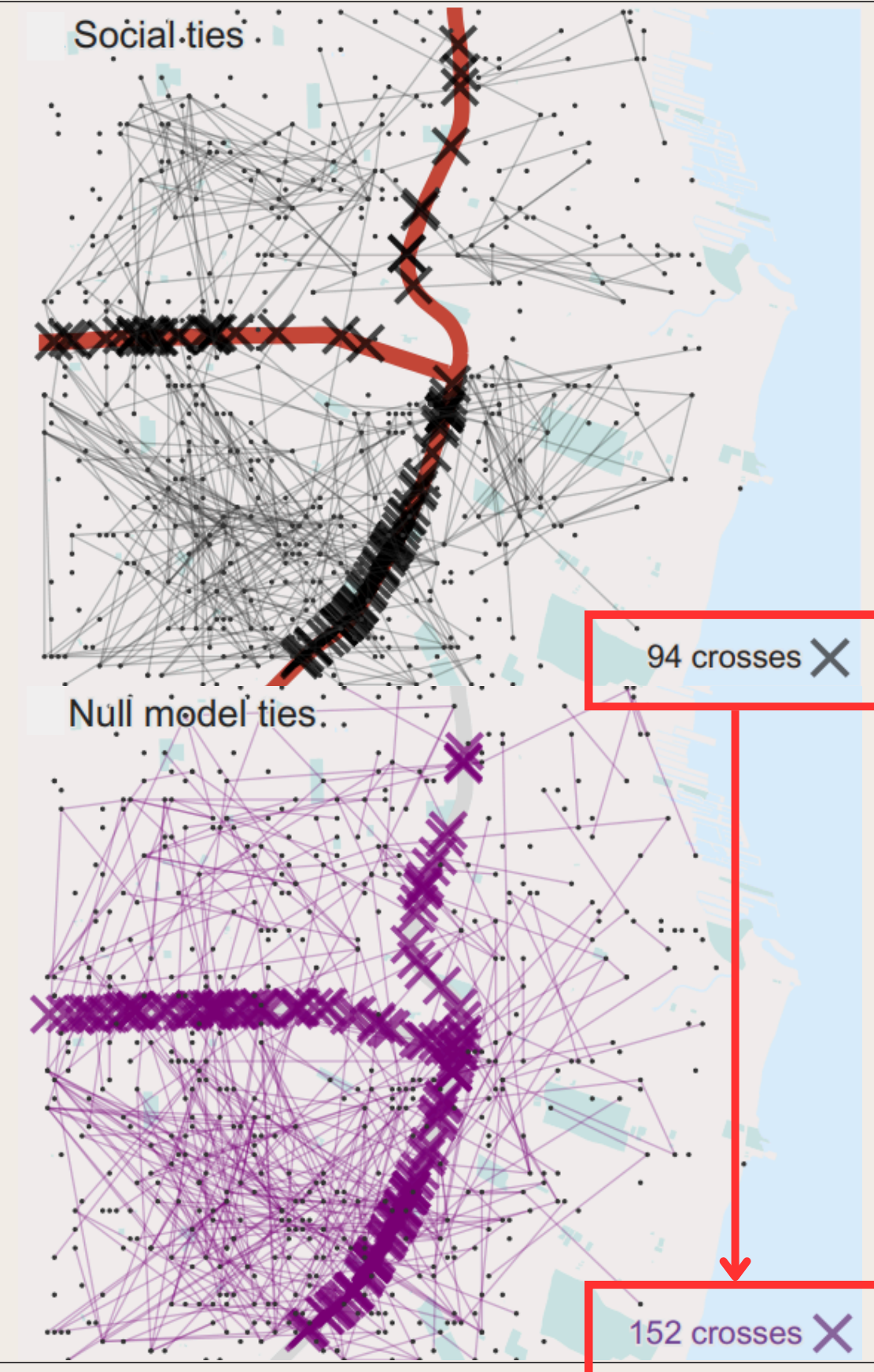
A measure of how much highways reduce social ties

High score = **stronger barrier effect**

Social ties = mutual connections between people

Compare observed vs. expected crossings

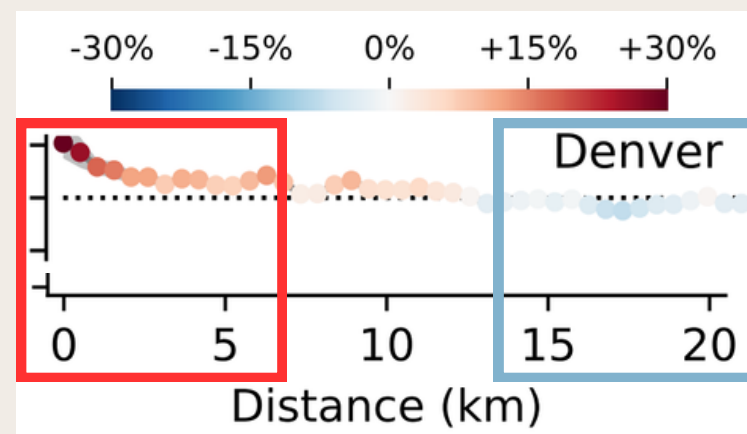
expected = randomized network (null model)



BARRIER SCORES RESULTS

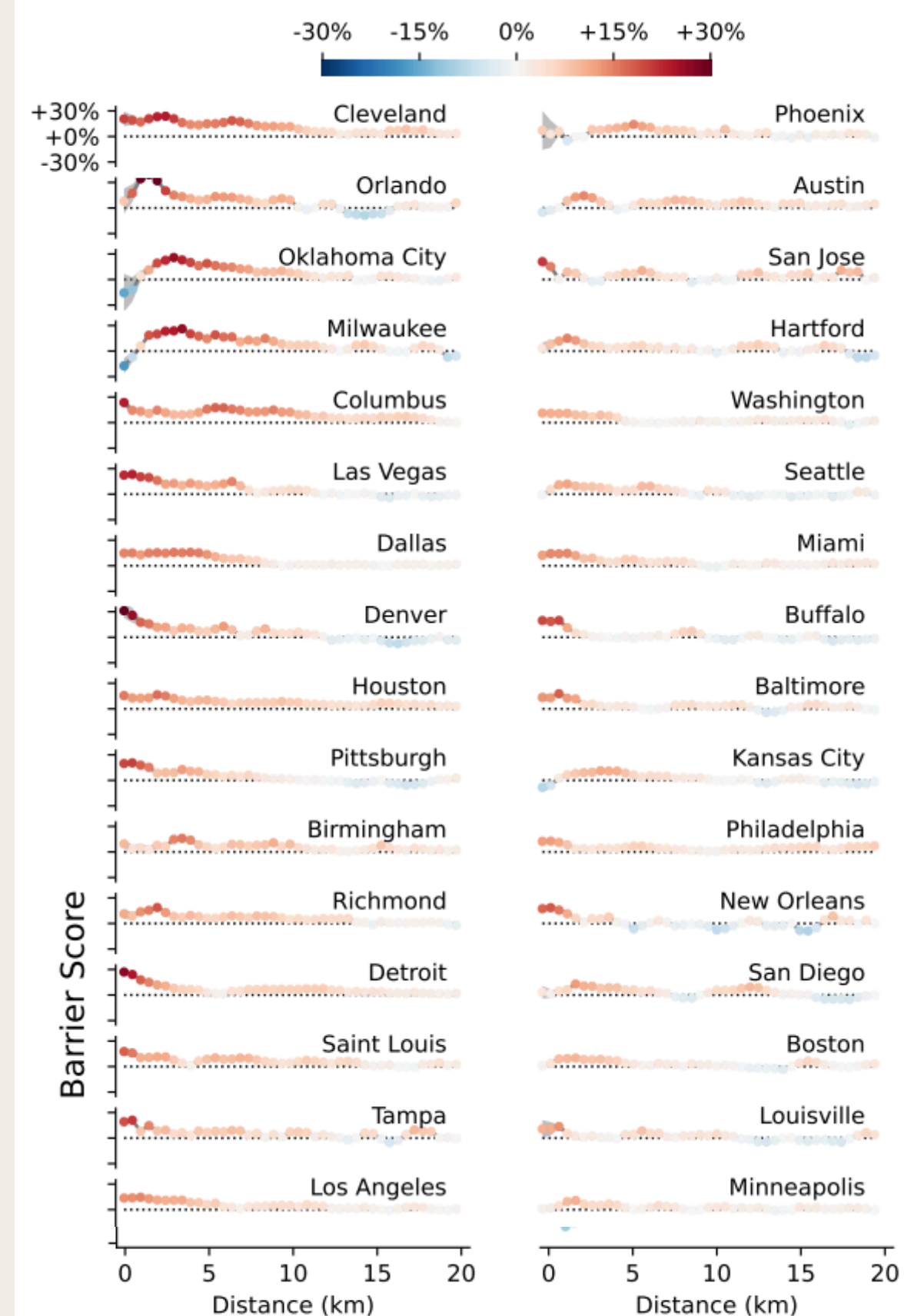
Positive in all 50 cities → highways consistently reduce social ties

Some cities show up to +16% reduction in ties



Strongest barrier effect at **short distances** (1-5 km)

→ Effect **fades** after ~20 km



SEGREGATION-ERA HIGHWAYS STILL SHAPE SOCIAL CONNECTIVITY



Highest barrier scores on segregation-era highways

These barriers persist today, decades after construction

Social connectivity drops exactly along historic racial boundaries

LIMITATIONS

Twitter mutual follows $\neq$ full social relationships

Not all follows are meaningful ties

Data captures online ties only

Offline relationships are missing
→ Underestimates real-world relationships

Null model simplifies real mobility patterns

Real mobility is more complex than random ties

Not causal: highways correlate with multiple factors

Highway is strongly associated with the divide, but can't say the highway caused it

TAKEAWAYS

Highways shape modern social networks

Positive scores in all 50 cities → highways
reduce ties everywhere

Barriers persist even without physical distance

Up to 16% fewer ties even at 1-5 km
distance

Historic segregation still influences present-day connectivity

Segregation-era highways show the
strongest barriers

Urban planning decisions have long-term social effects

Effects remain visible decades later

HIGHWAYS DIVIDE MORE THAN LAND. THEY DIVIDE PEOPLE.

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